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Smart Tourism Crowd Management System

نظام ذكي لإدارة الزحام في الأماكن السياحية

Project Proposal

Submitted by

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Abstract:

Tourist attractions in Egypt often experience significant overcrowding, especially during peak hours, leading to long waiting times and a poor visitor experience. This project proposes a smart crowd management and decision-support system that enhances the tourism experience by providing real-time crowd estimation, waiting time prediction, and intelligent visit planning.

The system is designed with a multi-layered data acquisition approach. First, it introduces a real-world sensor-based concept using entry and exit counting systems to track the number of visitors accurately. Second, a simulation layer is implemented using controlled random data generation to mimic realistic crowd behavior across different times of the day. Third, an advanced AI-based approach is considered using computer vision techniques (e.g., YOLO) to detect and count people through camera input.

By combining these approaches, the system ensures flexibility between real deployment and simulation. The application provides features such as crowd level monitoring, waiting time estimation, best time recommendations, alternative location suggestions, smart notifications, and full-day trip planning. The system aims to reduce waiting time, improve visitor satisfaction, and optimize tourism flow efficiently.

Problem:

Tourists visiting popular attractions in Egypt often face overcrowding and long queues, especially at entry gates. This results in wasted time, poor experience, and inefficient trip planning. Additionally, there is no intelligent system that provides real-time insights or predictive recommendations based on crowd conditions.

Solution:

The proposed system provides an intelligent solution through the following components:

1. Crowd Monitoring System:

Tracks and displays crowd levels (Low, Medium, High) and the number of people currently in the location.

2. Waiting Time Estimation:

Calculates expected waiting time based on queue size, number of service counters, and average service time.

3. Best Time Recommendation:

Suggests optimal visiting times based on crowd patterns and predicted data.

4. Smart Trip Planning:

Generates a full-day plan for tourists based on crowd levels across multiple locations.

5. Alternative Suggestions:

Recommends less crowded nearby locations when a place is highly congested.

6. What-if Analysis:

Allows users to simulate visiting at different times and compare waiting times.

7. Smart Notifications:

Notifies users when crowd levels become favorable.

Data Acquisition Approach (Key Innovation):

The system supports three integrated approaches for obtaining crowd data:

1. Sensor-Based Approach (Real-World Concept):

The system is designed to work with entry/exit sensors installed at gates. Each person crossing the gate triggers a signal.

- If entering → PeopleEntered++
- If exiting → PeopleExited++
- CurrentPeople = Entered – Exited

This provides accurate real-time tracking of crowd size.

2. Simulation Layer (Implementation):

Due to hardware limitations, the system simulates sensor behavior using controlled random number generation.

- Different time ranges (morning, afternoon, evening) have different patterns
- Entry and exit values are generated within realistic ranges
- Ensures non-negative and logical crowd flow

This creates a realistic approximation of real-world behavior.

3. AI-Based Detection (Advanced Extension):

The system can be extended using computer vision techniques such as YOLO to detect and count people using camera input.

This provides a scalable and automated solution without physical sensors.

System Features:

1. Real-time crowd monitoring with color-coded levels — Low, Medium, High, and Very High
2. Estimated waiting time calculation based on live crowd data and number of service counters
3. Best visiting time recommendation using hourly crowd forecasts for each attraction
4. What-If simulation that shows predicted crowd level and waiting time for any hour of the day
5. Smart trip planner that sorts selected places by crowd level and builds the most efficient route
6. Alternative attraction suggestions when a chosen place is too crowded
7. Virtual queue system with ticket numbers and estimated time slots

8. Smart push notifications when the tourist's turn is getting close
9. Premium booking for tours, tickets, and guided experiences
10. Automatic commission calculation on every booking
11. Payment processing with success and failure handling
12. Tourist profile with points, badges, and efficiency score
13. Wishlist to save favorite places
14. Reviews and ratings for each attraction
15. Admin dashboard with full system report and revenue tracking
16. Activity log that records every action in the system
17. Three-layer crowd data collection using physical sensors, AI camera, and data simulation

Benefits:

1. Tourists can plan their visit before leaving home and avoid peak hours
2. Waiting time at popular attractions is reduced through smart queue management
3. The overall tourist experience in Egypt is improved leading to better satisfaction and reviews
4. Attraction administrators get accurate real-time data and full reports without manual effort
5. The system works even without physical hardware thanks to the data simulator
6. New attractions can be added at any time without changing the system structure
7. New booking types can be added easily by creating a new subclass of Booking
8. All data is saved automatically so nothing is lost after closing the application
9. The commission system creates a sustainable revenue model for the platform
10. Gamification encourages tourists to use the app more and plan smarter trips